Linear Algebra for Data Analysis

In the Math syllabus, a linear equation was defined as:

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b; \sum_{i=1}^{n} a_ix_i = b$$

And a set of linear equations was defined as

$$a_{11}x_1 + a_2x_2 + \cdots + a_nx_n = b_1$$
$$\cdots$$
$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m$$

Solving for a set of equations:

Linear algebra was introduced to solve a set of linear equations, like the following example. Besides the equations we can also explore the system graphically, like in the figure.

$$\begin{aligned} 2x - 3y &= 2.5 \\ x + 4y &= 4 \end{aligned} \quad [1]$$

Usually, a set of two equations and two unknowns (x, y), can be solved and there is one solution where the lines meet, like in this example where the solution is $x = 2, y = 0.5$.

In matrix notation this problem is often written as $\boldsymbol{Ax} = \boldsymbol{b}$, with $\boldsymbol{A} = \begin{bmatrix} 2 & -3 \\ 1 & 4 \end{bmatrix}$, $\boldsymbol{b} = \begin{bmatrix} 2.5 \\ 4 \end{bmatrix}$, and $\boldsymbol{x}$ represents the vector of unknowns (in this example the combined unknowns called x and y) that have to be obtained.

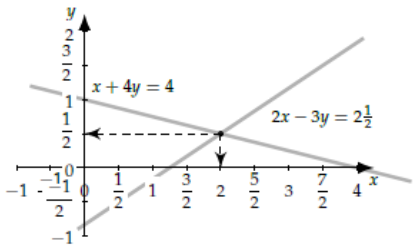


Figure 1: Graphical solution for a set of linear equations.

Such a set of linear equations can have a single solution, as in the example, it can also have no solution, and it can have an infinite number of solutions.

$\begin{aligned} x + y &= 2 \\ 2x + 2y &= 4 \end{aligned}$, will have an infinite number of solutions, but $\begin{aligned} x + y &= 2 \\ -x - y &= 3 \end{aligned}$ will have none.

If you make graphical representations of the systems, it will become clear why there are infinite or no solutions.

Some rules for notation:

In the BDA course we will use a different way of notation, which is more general in the data analysis literature. The measured data is defined as $\boldsymbol{X}$, with size (IxJ), in which $i = 1 \cdots I$, (rows of $\boldsymbol{X}$) represents the number of subjects (e.g. $I = 100$), and $j = 1 \cdots J$, (columns of $\boldsymbol{X}$) represents the number of metabolites (e.g. $J = 500$). The response we want to predict is indicated with $\boldsymbol{y}(Ix1)$, a vector, and the coefficients we want to estimate are indicated as $\boldsymbol{b}(Jx1)$. The matrix equation in BDA notation is thus $\boldsymbol{y} = \boldsymbol{Xb}$. Note that above, $\boldsymbol{X}$ was called $\boldsymbol{A}$, $\boldsymbol{y}$ was called $\boldsymbol{b}$, and $\boldsymbol{b}$ was called $\boldsymbol{x}$. Sorry for this confusion! From now on, we will use this new definition of notation.

If in all equations we use the same style for matrices, vectors and scalars, it makes understanding the equations much easier.

Matrices: Uppercase bold character: $\boldsymbol{X}(IxJ)$

Vectors: lowercase bold character: $\boldsymbol{y}(Ix1)$, or $\boldsymbol{b}(Jx1)$

Scalars: lowercase regular character: x_{ij}, y_i, b_j

Transpose and inverse: $\boldsymbol{X}^T, \boldsymbol{y}^T$ and $\boldsymbol{X}^{-1}$

Identity matrix: $\boldsymbol{I}$

Linear algebra for data analysis

For a set of equations, the matrix equation $\boldsymbol{Xb} = \boldsymbol{y}$, represents a problem where we have e.g. metabolite levels of 500 metabolites measured from plasma samples of 100 subjects of which their BMI is also available. To estimate which metabolites are related to the BMI response, regression coefficients $\boldsymbol{b}$ are calculated, and the ones which are different from 0 have an effect on the BMI. Here $\boldsymbol{X}(100 \times 500)$ represents the data with 100 rows and 500 columns. Each row contains information of 500 metabolites for a single subject, while each column represents the levels of a single metabolite for each individual. The BMI is represented by the vector $\boldsymbol{y}(100 \times 1)$, while the regression coefficients are stored in vector $\boldsymbol{b}(500 \times 1)$.

Infinite number of solutions

In the metabolome system described above, there are more metabolites than subjects, $\boldsymbol{X}$ has more columns than rows, or there are more unknown parameters to estimate than the number of equations. Such systems generally have an infinite number of solutions.

As an example, take the first set of linear equation [1]. If we replace x and y with b_1 and b_2 and we add an unknown b_3.

$$\begin{aligned} 2b_1 - 3b_2 + b_3 &= 2.5 \\ b_1 + 4b_2 - b_3 &= 4 \end{aligned} \qquad [2]$$

The solution $b_1 = 2, b_2 = 0.5, b_3 = 0$ was already found earlier, but also the solution $b_1 = 0, b_2 = 6.5, b_3 = 22$ is valid, and there are many more. With many solutions we have no idea which metabolites have an effect on BMI, and thus this situation is not useful.

No solutions

The situation that there are no solutions to solve the set of equations is often obtained when more equations than unknowns are present in the system, or $\boldsymbol{X}$ has more rows than columns.

Let's take the system

$$\begin{aligned} 2b_1 - 3b_2 &= 2.5 \\ b_1 + 4b_2 &= 4 \\ b_1 + b_2 &= 1 \end{aligned} \qquad [3]$$

This system has three lines that do not all go through a single point, and thus there is no solution for b_1 and b_2 that exactly provide the $\boldsymbol{y}$ response. In such cases we look for the solution of b_1 and b_2 for which $\boldsymbol{Xb} = \hat{\boldsymbol{y}}$ is as close as possible to $\boldsymbol{y}$. Here, when using the least squares approach, close means that the sum of the squared difference between $\hat{\boldsymbol{y}}$ and $\boldsymbol{y}$ is as small as possible. This is $b_1 \approx 1.74, b_2 \approx 0.43$. This is a useful solution. Although multiplying $\boldsymbol{Xb}$ does not provide $\boldsymbol{y}$ exactly, it is the closest we can get. Realizing that all measurements in $\boldsymbol{X}$ and $\boldsymbol{y}$ come with errors but knowing that they are close to their true values, the estimated values for b_1 and b_2 also have small errors, but are still close to their true values. Because the system cannot be solved exactly, the matrix equation is written as $\boldsymbol{y} = \boldsymbol{Xb} + \boldsymbol{e}$, where $\boldsymbol{e}$ is a vector of the same size as $\boldsymbol{y}$ and represents the residual between the obtained solution $\boldsymbol{Xb}$ and $\boldsymbol{y}$.

Concluding: systems with more unknowns than equations, i.e. $(\boldsymbol{X})$ has more columns than rows usually, lead to an infinite number of solutions, which is problematic, while systems with more equations than unknowns where $(\boldsymbol{X})$ has more rows than columns, lead to no solutions, but still give us a useful approximate solution.

Another graphical representation

In Figure 1, each line represents an equation, or each line represents one row of the $\boldsymbol{X}$ matrix. The number of dimensions (axes x and y) in Figure 1 is defined by the number of columns in $\boldsymbol{X}$. Another way of visualizing the set of equations is by plotting each column of $\boldsymbol{X}$ as a vector and also plotting $\boldsymbol{y}$ as a vector. The dimension (number of axes) of this plot equals the number of rows in $\boldsymbol{X}$, which also equals the number of values in $\boldsymbol{y}$.

This is the column space representation of matrix $\boldsymbol{X}$ from equation [3]. $\mathbf{x}_1$ and $\mathbf{x}_2$ represent the first and second column of $\boldsymbol{X}$ and $\boldsymbol{y}$ is the third vector.

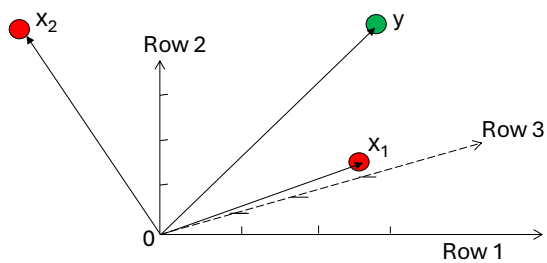


Figure 2: Column space representation of a matrix.

Vector multiplication

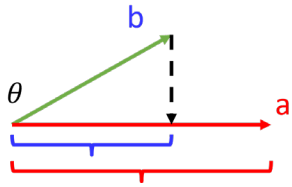
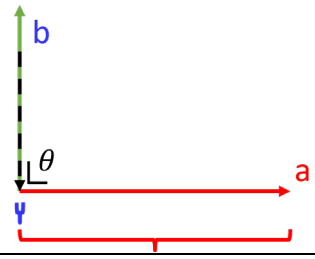
A very important concept in linear algebra for data analysis is the multiplication of two vectors. This can be done in two ways: the inner product that provides a scalar: $c = \boldsymbol{a}^T\boldsymbol{b}$, and the outer product that provides a matrix: $\boldsymbol{C} = \boldsymbol{ab}^T$, where $\boldsymbol{a}$ and $\boldsymbol{b}$ are both column vectors of the same size.

Inner product

$$\mathbf{a} \cdot \mathbf{b} = \boldsymbol{a}^T\boldsymbol{b} = \sum_{i=1}^{I} a_i b_i \qquad [4]$$

From the equation you see that both vectors need to have the same number of elements (I) and that the result is a scalar. This is sometimes called the dot product ($\mathbf{a} \cdot \mathbf{b}$) or the inner product of two vectors. The inner product is also equal to the length of $\mathbf{a}$ times the length of $\boldsymbol{b}$, times the cosine of their angle. The inner product is a measure of how much 2 vectors go in the same

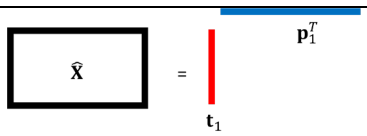
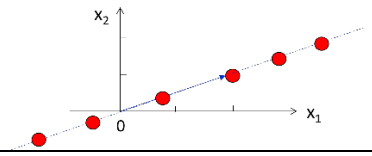
direction. If they go in orthogonal directions, where the angle is 90 degrees, the inner product is 0.

$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \cdot \|\mathbf{b}\| \cdot cos(\theta)$ $\|\mathbf{a}\| = \sqrt{a_1^2 + a_2^2 + \cdots a_I^2}$		

The inner product is also used when multiplying two matrices $\boldsymbol{C} = \boldsymbol{AB}$. Each row of $\boldsymbol{A}$ is multiplied with each column of $\boldsymbol{B}$ using the inner product approach and provides an element of $\boldsymbol{C}$. This can only be done when the row of $\boldsymbol{A}$ and the column of $\boldsymbol{B}$ have the same number of values. $\boldsymbol{C}$ then automatically will have the same number of rows as $\boldsymbol{A}$ and the same number of columns as $\boldsymbol{B}$.

Outer product:

The outer product makes a matrix out of two vectors

$\widehat{\boldsymbol{X}} = \boldsymbol{t}_1 \boldsymbol{p}_1^T$		

In this example related to Principal Component Analysis, where $\boldsymbol{t}_1$ is a vector providing information on samples and $\boldsymbol{p}_1$ is a vector with information on the variables, their product provides information on the samples in (in this example) two dimensions ($X_1$and X_2) if $\boldsymbol{p}_1$ only contains two values. However, you can also see that actually the information in this matrix is 1 dimensional in a two-dimensional space (as all points lie on the same line).

Actually, the outer product is also an inner product but for vectors of size 1, as the topleft element of $\widehat{\boldsymbol{X}}$ is the inner product of the first element of $\boldsymbol{t}_1$ and the first element of $\boldsymbol{p}_1$.

Exercise:

Give an example of two vectors that are orthogonal and show this mathematically and graphically.

Subspace, Null space, rank of a matrix

In the math syllabus the concepts of Null space, subspace and rank have been introduced. The null space was defined as the set of all solutions $\boldsymbol{b}$ for which $\boldsymbol{Xb} = \boldsymbol{0}$. Going back to Figure 2, where two column vectors of $\boldsymbol{X}$ were placed in a three-dimensional column space, the two vectors together span a two-dimensional subspace in a three-dimensional space. Any position in the subspace can be reached by $b_1\boldsymbol{x}_1 + b_2\boldsymbol{x}_2$, but the direction orthogonal to this subspace cannot be reached by these two columns. This not reachable part of the three-dimensional

space is called the null space. Any vector that has a contribution of this null space, cannot perfectly be described by $\boldsymbol{X}$. This is also the reason that of the residual $\boldsymbol{e}$ in $\boldsymbol{y} = \boldsymbol{Xb} + \boldsymbol{e}$. $\boldsymbol{e}$ is thus also a vector in the null space of $\boldsymbol{X}$, as $\boldsymbol{X}^T\boldsymbol{e} = \boldsymbol{0}$.

Similarly, a matrix $\boldsymbol{X}(IxJ)$ with more columns than $(I < J)$ rows can be represented as I vectors in the J-dimensional space. Because the J dimensional space is not fully covered there exist a null space in the rowspace of $\boldsymbol{X}$. Any vector that partly occupies this null space cannot be perfectly represented by the I directions in this space. The part that cannot be not covered lies in the null space of the row space of $\boldsymbol{X}$.

Projections

For a skinny data matrix $\boldsymbol{X}(IxJ)$ $(I > J)$ the column space had dimension I, but the rank is only J. This means that not all dimensions of the column space of $\boldsymbol{X}$ are occupied. Looking back at Figure 2, a $(3x2)$ matrix with two $(\boldsymbol{X})$ columns is represented as well as the vector $(\boldsymbol{y})$. The two $(\boldsymbol{X})$ vectors together describe a two-dimensional subspace inside the three-dimensional column space of $(\boldsymbol{X})$. Any position in that space can be obtained by a linear combination of the two vectors. Chances are that $\boldsymbol{y}$ lies outside the subspace of $\boldsymbol{X}$. The optimal solution to predict $\boldsymbol{y}$ from $\boldsymbol{X}$ is to orthogonally project $\boldsymbol{y}$ on the subspace. The orthogonal projection makes that the distance between $\boldsymbol{y}$ and its projection is as small as possible.

The projection matrix $\boldsymbol{H}$ can be obtained as $\boldsymbol{X}(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T$ and thus the projection is obtained as $\hat{\boldsymbol{y}} = \boldsymbol{Hy}$. One of the properties of the projection matrix is that it is **idempotent**. This means $\boldsymbol{HH} = \boldsymbol{H}$. If the projected $\hat{\boldsymbol{y}}$ is projected again on the subspace, then this does not have an effect, as it is already in the subspace.

Exercise:

Show that the matrix $\boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T$ is idempotent.

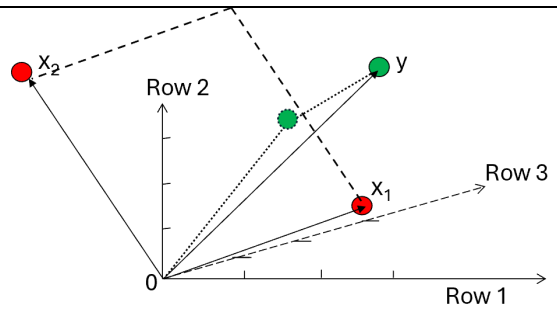


*Figure 3: Projection of **y** onto space defined by $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$.*

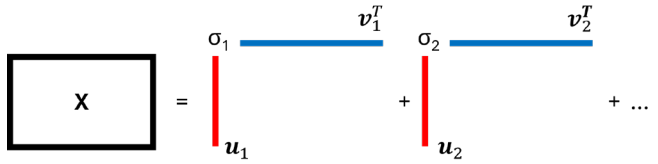
Inverse of a matrix

The product of the inverse of a matrix times the matrix itself produces the identity matrix: $\boldsymbol{XX}^{-1} = \boldsymbol{X}^{-1}\boldsymbol{X} = \boldsymbol{I}$. Only square matrices can have an inverse if that matrix is of full rank. In the math syllabus it was shown how to calculate inverse of a matrix using Gaussian elimination or Gauss-Jordan elimination. In solving a set of linear equations: $\boldsymbol{y} = \boldsymbol{Xb}$, then $\boldsymbol{b} = \boldsymbol{X}^{-1}\boldsymbol{y}$, if $\boldsymbol{X}$ is a square matrix and $\boldsymbol{X}^{-1}$ exists.

In many cases in regression analysis, $\boldsymbol{X}$ will not be square. In that case the solution $\boldsymbol{b} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y}$ can be used. This $(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T$ is called the Least Squares inverse. Note that for $\boldsymbol{X}$ of size (IxJ) with $(I > J)$, $(\boldsymbol{X}^T\boldsymbol{X})$ will have size (JxJ), and will have rank J, so the inverse will exist. For $\boldsymbol{X}$ with $(I < J)$, $(\boldsymbol{X}^T\boldsymbol{X})$ will also have size (JxJ), but will only have rank I, and thus is not of full rank. In that case the inverse does not exist, which means that in this case the regression coefficients $\boldsymbol{b}$ cannot be estimated in this way, and alternative methods have to be used.

Singular Value Decomposition (SVD).

The singular value decomposition aims to describe a matrix as a sum of outer products of column (left) singular vectors and row (right) singular vectors. Singular vectors have special properties.

The SVD of $\boldsymbol{X}$ provides left singular vectors $\boldsymbol{u}$ and right singular vector $\boldsymbol{v}$ and singular values $\boldsymbol{\sigma}$.	

Singular vectors $\boldsymbol{t}_1$ and $\boldsymbol{p}_1$ have the special property that if $\boldsymbol{X}\boldsymbol{v}_1 = \sigma\boldsymbol{u}_1$ then $\boldsymbol{X}^T\boldsymbol{u}_1 = \sigma\boldsymbol{v}_1$, with σ being the singular value. All left singular vectors are orthogonal to each other and also all right singular vectors are orthogonal to each other. In matrix notation the singular value decomposition is written as:

$$\boldsymbol{X} = \boldsymbol{U}\boldsymbol{S}\boldsymbol{V}^T$$

With $\boldsymbol{U}(IxI)$; $\boldsymbol{U}^{\boldsymbol{T}}\boldsymbol{U} = \boldsymbol{U}\boldsymbol{U}^{\boldsymbol{T}} = \boldsymbol{I}_I$ and $\boldsymbol{V}(JxJ)$; $\boldsymbol{V}^{\boldsymbol{T}}\boldsymbol{V} = \boldsymbol{V}\boldsymbol{V}^{\boldsymbol{T}} = \boldsymbol{I}_J$ and $\boldsymbol{S}(IxJ)$ has an upper left block with singular values on its diagonal and zeros anywhere else. $\boldsymbol{I}_I$ represents the identity matrix of size IxI, and $\boldsymbol{I}_J$ represents the identity matrix of size JxJ. See next example.

$$\begin{bmatrix} 2 & 3 & 4 \\ 5 & 3 & 1 \\ 1 & 3 & 2 \\ 4 & 6 & 8 \end{bmatrix} = \begin{bmatrix} -0.40 & -0.16 & 0.12 & -0.89 \\ -0.35 & 0.93 & 0.05 & 0.00 \\ -0.27 & -0.05 & -0.96 & 0.00 \\ -0.80 & -0.32 & 0.24 & 0.45 \end{bmatrix} * \begin{bmatrix} 13.34 & 0 & 0 \\ 0 & 3.84 & 0 \\ 0 & 0 & 1.18 \\ 0 & 0 & 0 \end{bmatrix} * \begin{bmatrix} -0.45 & 0.79 & 0.41 \\ -0.59 & 0.07 & -0.80 \\ -0.67 & -0.60 & 0.43 \end{bmatrix}^T$$

Often the SVD is used to remove noise from the data matrix by using only the singular vectors related to the largest singular values. This is called a truncated SVD. By using only the first two singular vectors, $\boldsymbol{X}$ can be approximated by:

$$\widehat{\boldsymbol{X}} = \begin{bmatrix} -0.40 & -0.16 \\ -0.35 & 0.93 \\ -0.27 & -0.05 \\ -0.80 & -0.32 \end{bmatrix} * \begin{bmatrix} 13.34 & 0 \\ 0 & 3.84 \end{bmatrix} * \begin{bmatrix} -0.45 & 0.79 \\ -0.59 & 0.07 \\ -0.67 & -0.60 \end{bmatrix}^T$$